MICHAEL W. HOPWOOD

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OVERVIEW

- Machine Learning Scientist with 6+ years of rich applied ML experience across FAANG and Research Labs
- Built and productionized agentic solutions and large-scale ML solutions across high and low-latency applications, impacting hundreds of millions of users and business decisions
- Founded 2 startup companies, which accrued $150k+ in VC seeding
- Published ML researcher with 140+ citations across deep learning, graph neural networks, probabilistic modeling, and ML for physical systems; Research during PhD funded by a National Laboratory

TECHNICAL SKILLS

Languages & Infrastructure	Python, SQL, C#, C, R, Azure ML, AWS, GCP, Azure, PyTorch, Jax, Tensorflow
Generative AI	LLMs, SLMs, distillation, DeepSpeed, FlashAttention, agentic systems, model distillation, LangGraph, CrewAI, Strands, AgentCore, Onnx, Huggingface
Machine learning	Deep tree models, transformers, graph neural networks, ensembles, active learning, hyperparameter tuning, one class learning
Data Engineering	Data integrity, data processing, pragmatic statistics reports
Software Development	CI/CD pipelines, unit tests, git

EXPERIENCE

AMAZON Dec.2025 – Present

Applied Scientist II, Agentic Guidance for Advertisers *Seattle, WA*

- Owned and shipped complex agentic solutions for providing guidance to advertisers in their ad campaign creation and optimization, resulting in improved sales and Return on Ad Spend (ROAS)
- Built five ML economic forecast models for predicting the effect of making an ad campaign edit, which feeds into all the team's agentic solutions; Process completed end to end, including monitoring dashboards

MICROSOFT Jul.2023 – Dec.2025

Applied Scientist II, Bing Search (Core Search, AI Team) *Seattle, WA*

- **Deep Search Squads:** *Led cross-functional teams* and served as lead developer in building low-latency distilled *small language models* (SLMs) and LLMs to improve Bing's core search algorithm, leading to the first completely online deployment of an SLM within the core search algorithm and rendered significantly improved search results
- **GenAI Query Understanding:** Trained SLMs with *customized loss functions* producing a *GenAI* query understanding guideline which was optimized for both metrics and literacy; introducing this signal into downstream models led to a dramatic +1.1% *relevance improvement*
- **Innerloop Model Validation:** Iterating on official metrics was slow leading to poor agility in model validation, so I built *high-fidelity innerloop* sets with proved End-to-End metric correlation, improving the agility of all members, rendering +50% faster shipment rates and fewer poor quality online *AB tests*
- **OpenAI Model Integration:** Worked with *OpenAI* folks to establish *high-throughput environments* for applying prompted LLMs in an asynchronous state on Bing Search; applied those models to improving the core search algorithm, which led to -2% defect rates, dramatically improving the cluttering issue on Bing search
- **Query Index Extension:** To widen the impact of expensive and slow LLMs, I designed a scalable *semantic-based cache lookup infrastructure* that applied similar cached results to a query, widening the LLM impact by >40% at small precision expense and leading to +0.6% *relevance gains*
- **Stack Simplification:** Removed tech debt by simplifying the ML model ecosystem within the search algorithm, including shipped models with overlapping scopes as well as legacy logic trees, leading to improved debuggability and agility in the *AB testing* of new candidate models, as well as +0.2% relevance due to removing antiquated models

AMAZON Aug.2022 – Nov.2022

PhD Applied Scientist Intern *Seattle, WA*

- Designed & trained novel *graph neural network* models with goal of reducing the amount of abuse on amazon.com
- Released internal package for training and productionizing customized graph neural networks

MICROSOFT May.2022 – Aug.2022

PhD Applied Scientist Intern *Seattle, WA*

- Designed and productionized ML models trained using *custom loss functions* for Bing search optimization, specifically focused on optimizing rendered page layout of search result
- Model results impact millions of people daily and achieved 2.4% *revenue improvements* (massive!)

TESLA Jan.2022 – May.2022

PhD Applied Scientist Intern *Palo Alto, CA*

- Designed and built systems towards interfacing supercharger charge demand signals – including building out forecasts - into the *Tesla trip planner*
- Built supercharger spacing models incorporating graph theory – achieved over 20% better performance than previous production models, translating to *millions of dollars saved* for the company in its infrastructure planning efforts

SANDIA NATIONAL LABORATORIES Aug.2020 – Jan.2022

PhD *Applied Scientist Intern* *Albuquerque, NM*

- Answered failure classification tasks via *customized deep learning*, physics-based simulations, and NLP; CICD GitHub project management setup and maintenance
- Published papers, gave conference presentations, and released an open-source python package

DATA SCIENCE DEPARTMENT, UCF Aug.2020 – Jan.2022

PhD Research Assistant *Orlando, FL*

- Designed *graph neural networks* for applications on social networks, power systems, and quantum mechanics; Advised undergraduate statistics & computer science students on data science
- Published papers and gave conference presentations

REVOLUTION MEDICINE Nov.2020 – Dec.2020

Data Scientist, Contractor *Gainesville, FL*

- Developed *ML assistant to aid doctors* with deducing best pharmaceutical intervention for a patient given genome and demographic using peer-review journal papers
- Product rolled out alongside the startup's hardware product

FLORIDA SOLAR ENERGY CENTER Oct.2017 – Jul.2020

Research Assistant *Orlando, FL*

- Utilized ML to detect and classify failures in solar fields; held quarterly Department of Energy stakeholder meetings
- *Data engineering* tasks to securely channel data across multiple networks without any loss of data

QUIRK TECHNOLOGIES, LLC Feb.2019 – Mar.2020

Engineer, Contractor *Orlando, FL*

- Developed business growth and development recommendation via customized *business analytics* pipeline
- Designed 3D models for a manufacture-grade point-of-sales product

OSISOFT, LLC May.2018 – Aug.2018

Academic Intern *Philadelphia, PA*

- Generated python API which interfaced to a proprietary archive with a *.NET backbone*
- Ensured the health of a *real-time data management* infrastructure by monitoring the flow of data across platforms

ENGINEERING DEPARTMENT, UCF Oct.2017 – Mar.2018

Undergraduate Research Assistant *Orlando, FL*

- Studied the effects of modular defects on solar cells using support vector machines; accumulated and archived all failures discovered in solar cells to date

PHYSICS DEPARTMENT, UCF Aug.2017 – Dec.2017

Physics Teaching Assistant *Orlando, FL*

- *Prepared and taught lectures* to 90+ students; held office hours and exam reviews

SCIENTIFIC JOURNAL PUBLICATIONS (check scholar for latest)

- Bonney, Kirk Layne, **et al.** "pvOps: a Python package for empirical analysis of photovoltaic field data." *Journal of Open Source Software* 8.SAND-2023-13746J (2023). https://www.osti.gov/biblio/2311477
- **Hopwood, Michael.** *One-class systems seamlessly fit in the forward-forward algorithm.* Arxiv (2023). https://arxiv.org/abs/2306.15188

- **Hopwood, Michael,** Lekha Patel, and Thushara Gunda. *Classification of Photovoltaic Failures with Hidden Markov Modeling, an Unsupervised Statistical Approach*. Energies 15.14 (2022): 5104. https://doi.org/10.3390/en15145104
- **Hopwood, Michael,** et al. *Physics-Based Method for Generating Fully Synthetic IV Curve Training Datasets for Machine Learning Classification of PV Failures*. Energies 15.14 (2022): 5085. https://doi.org/10.3390/en15145085
- **Hopwood, Michael,** and Thushara Gunda. *Generation of Data-Driven Expected Energy Models for Photovoltaic Systems*. Applied Sciences 12.4 (2022): 1872. https://doi.org/10.3390/app12041872
- Mantzaris, Alexander V., **et al.** *Tagasaurus, a tool to assist manual image tagging and the creation of image collections.* Software Impacts 10 2021: 100157. https://doi.org/10.1016/j.simpa.2021.100157
- Mantzaris, Alexander V., **et al**. *Introducing tagasaurus, an approach to reduce cognitive fatigue from long-term interface usage when storing descriptions and impressions from photographs.* Technologies 9.3 (2021): 45. https://www.mdpi.com/2227-7080/9/3/45
- **Hopwood, Michael**, et al. *Exploring the value of nodes with multicommunity membership for classification with graph convolutional neural networks*. Information 12.4 (2021): 170. https://www.mdpi.com/2078-2489/12/4/170
- **Hopwood, Michael**, Phuong Pho, and Alexander Mantzaris. *Exploring the strength of weak ties to direct active learning for community label prediction*. No. SAND2021-4095C. Sandia National Lab.(SNL-NM), Albuquerque, NM (United States), 2021. https://www.osti.gov/servlets/purl/1862776
- **Hopwood, Michael**, et al. *Neural network-based classification of string-level IV curves from physically-induced failures of photovoltaic modules*. IEEE Access 8 (2020): 161480-161487. https://ieeexplore.ieee.org/abstract/document/9186596

SCIENTIFIC PRESENTATIONS AND PUBLISHED PROCEEDINGS (check scholar for latest)

- **(Panelist) Hopwood, Michael**, Lekha Patel, and Thushara Gunda. *Probabilistic detection of high-dimension failures in complex systems: A case study of photovoltaics*. No. SAND2021-10806C. Sandia National Lab.(SNL-NM), Albuquerque, NM (United States), October 2021. INFORMS Annual Meeting, Anaheim, California https://www.osti.gov/servlets/purl/1886187
- **(Oral) Hopwood, Michael**, Phuong Pho, and Alexander V. Mantzaris. *Exploring a link between network topology and active learning*. 2021 Twelfth International Conference on Ubiquitous and Future Networks (ICUFN). IEEE, 2021. (Published proceedings) https://ieeexplore.ieee.org/abstract/document/9528662
- **(Oral)** Mendoza, Hector, **Michael Hopwood**, and Thushara Gunda. *pvOps: Improving operational assessments through data fusion*. 2021 IEEE 48th Photovoltaic Specialists Conference (PVSC). IEEE, 2021. https://www.osti.gov/servlets/purl/1872512
- **(Oral) Hopwood, Michael**, Hector Mendoza, and Thushara Gunda. *Generating actionable information through the fusion of text and timeseries data: A case study of extreme weather effects at Photovoltaic plants*. No. SAND2020-12816C. Sandia National Lab.(SNL-NM), Albuquerque, NM (United States), 2020. American Geophysical Union, Fall Meeting 2020. https://ui.adsabs.harvard.edu/abs/2020AGUFMIN0140003H/abstract
- **(Oral, Best Paper Award) Hopwood, Michael,** et al. *An assessment of the value of principal component analysis for photovoltaic IV trace classification of physically-induced failures*. 2020 47th IEEE Photovoltaic Specialists Conference (PVSC). IEEE, 2020. https://ieeexplore.ieee.org/abstract/document/9300601
- **(Poster)** Gabor, Andrew M., **et al.** *The impact of cracked solar cells on solar panel energy delivery.* 2020 47th IEEE Photovoltaic Specialists Conference (PVSC). IEEE, 2020. https://ieeexplore.ieee.org/abstract/document/9300743
- **(Poster)** Walters, Joseph, **et al.** *Experimental methods to replicate power loss of pv modules in the field for the purpose of fault detection algorithm development*. 2019 IEEE 46th Photovoltaic Specialists Conference (PVSC). IEEE, 2019. https://ieeexplore.ieee.org/abstract/document/8980896

EDUCATION

2020-2023	**DATA SCIENCE & STATISTICS (M.S.)**	**University of Central Florida**
	Dissertation broadly in Graph Neural Networks for Physics	
2020	**MECHANICAL ENGINEERING (B.S.)**	**University of Central Florida**
	Burnett Honors College, Mathematics Minor	

PRESTIGIOUS AWARDS

- **OUC ML Competition, 2021** runner up award for temporal energy modeling with cash prize
- **Best Student Presentation Award** at PVSC47, an international conference, in "Solar Resource for PV and Forecasting"
- Honorable mention in international Mathematical Contest in Modeling 2020
- UCF's Gold Pegasus scholarship, 2016